

ZOO 507

Important M.C.S Midterm

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Lecture no 01

1-An interdisciplinary study of how humans interact with the environment of living and nonliving things is called the
Environmental science.

Lecture No :02

1-Study of the Earth water resources is called the**hydrology.**

2-Study of the organism interact with one another & with their non-living environment is called the **Ecology.**

3- Study of the Earth as whole & its non-living system is called**Earth science.**

4-Environmental science is a study of connections between the earth's life-support system and human activities..... **Interdisciplinary.**

LECTURE NO 03

1-A key subfield of environmental science is..... **ecology.**

2- A set of organisms interacting with one another and with their environment of nonliving matter and energy within a defined area or volume is called the**Ecosystem.**

LECTURE NO 04

1-Sustainability is the ability of the earth's various.....**natural system & human culture system.**

2- A critical component of sustainability is..... natural capital(the natural resources and natural services).

LECTURE NO 05

1-The resources or the materials and energy in nature that are essential or useful to humans are called the Natural resources.

2-The natural resources are classified into how many types.....(2).

3-One of the vital natural service is known as the nutrient cycling.

LECTURE NO 06

1-Natural capital is supported by the solar capital.

LECTURE NO 10

1-Economic growth is an increase in a nation's output of goods and services.

2-The annual market value of all goods and services produced by all firms and organizations, foreign and domestic, operating within a country is called the GDP(gross domestic product).

3- To help compare countries, economists use a tool called..... PPP(purchasing power parity).

LECTURE NO 13

1-The management of natural resources with the goal of minimizing resource waste and sustaining resource supplies for current and future generations is called the..... conservation.

LECTURE NO 14

1-Solar energy is called a.....resource because it is renewed continuously and is expected to last at least years as the sun completes its life cycle..... (perpetual resource ,6 billion years).

2 -The highest rate at which a renewable resource can be used indefinitely without reducing its available supply is called it'ssustainable yield.

3- When we exceed a renewable resource's natural replacement rate, the available supply begins to shrink, a process known asenvironmental degradation.

LECTURE NO 15

1-There are how many types of the property of the resource light.....three(3).

LECTURE NO 18

1-The process which involves collecting waste materials and processing them into new materials is called therecycling.

LECTURE NO 19

1-In 2003, humanity total or global ecological footprint was about than the earth ecological capacity.....25% higher.

LECTURE NO 20

1-After the oil-rich United Arab Emirates, the United States has the world's largest per capita ecological footprintSecond.

LECTURE NO 22

1-The word 'ecology' was first used by Ernest Haeckel in1869.

2- Ecology is a complex and an amazing field of biology that describes of the biosphere.....Richness.

3- This is the basic concept of, “all organisms in an environment are interdependent”Ecology.

4-How many major components of the ecology are the.....6.

LECTURE NO 23

1-The word ecosystem was coined by Tansley in..... 1935.

2- Ecosystem is a basic structural and functional unit ofecology.

3- Structure of an ecosystem depends onspecies diversit .

4 An ecosystem consists of how major components2(biotic& a biotic).

5-Biotic components of an ecosystem includes..... Producers & consumers.

6-A biotic components of the ecosystem includes..... Decomposers & physical components & chemical components.

LECTURE NO 24

1-A positive feedback loop causes a system to change further in the form which it is moving.....Same direction.

2-A feedback loop causes a system to change in the opposite direction from which it is moving is called..... Negative or the corrective feedback.

3- The ability of a living system, such as grassland or a forest, to survive in conditions of moderate disturbances is called..... Inertia or persistence.

4- The ability of a living system to be restored through secondary succession after a moderate disturbance is called..... Resilience.

LECTURE NO 25

1-Cell is the basic structural and functional unit of..... life.

2- All organisms are composed of cells which is the basic concept of..... cell theory.

3- Organisms may consist of..... Single or the multicellular.

4- Organisms are divided into two types on the basis of theircell structure.

LECTURE NO 26

1-A group of sexually reproducing organisms that can mate and produce fertile offspring is called.....species.

2- The species is the largest unit of in which gene flow is possible.....population.

3-How many concepts are about the species.....4

4-The two species are in the same area but live in different habitats is called the.....ecological species concept.

LECTURE NO 27

1-A group of same species located at the same place and same time is called as a..... Population.

2- Geographic limits are also a main constraints that decreases the size of a..... **population.**

LECTURE NO 28

1-The group of population of different species located in a particular area or place is called **community or the biological community.**

LECTURE NO 30

1-The ecological niche is an important concept of **community structure.**

LECTURE NO 31

1-Those species that have broad range of niches are known as **generalist species.**

2- Cockroaches are known as the **Nature's Ultimate Survivors.**

3- Cockroaches are present on earth since **350 million years.**

4- One species of cockroaches can survive being frozen for **48 hours.**

5- In only a year, a single Asian cockroach and its offspring can add about new cockroaches to the world..... **10 million.**

6- About species of cockroach live in homes and can carry viruses and bacteria that cause diseases..... **25.**

LECTURE NO 32

1-Those species that occupy narrow range of niches are known as..... **specialist species.**

2- China's giant panda is highly endangered because of a combination of habitat loss.....Endangered.

LECTURE NO 33

1-Those species that normally live and thrive in a particular ecosystem because of its soil, isolation and climatic conditions are called the.....native species.

2-Native species can be divided into the2 types.

3- These species are found within a specific part of world only.....Endemic species.

4- These species are found within the region and elsewhere in the world also.....Indigenous species.

5- Blind Indus dolphin is only found in Indus river ofPakistan.

6- Red kangaroo is endemic to only in the World.....Australia.

LECTURE NO 34

1-Species that are non-native to their environment and are deliberately or accidentally introduced into an ecosystem are callednon-native species.

2- Grass carp was introduced in Pakistan fromchina.

LECTURE NO 35

1-Species that provide early warnings of damage to a community or an ecosystem are called indicatorspecies.

2- Birds are excellent because they are found almost everywhere.....Biological indicators.

LECTURE NO 36

1-Some species play a key and crucial role in maintaining the structure of a biological community. They are called..... **Keystone species.**

2- The loss of a keystone species can lead to..... **population** crashes.

LECTURE NO 37

1-The species plays a major role in shaping communities are called the..... **foundation species.**

2-Beavers are acting as the **ecological engineers.**

LECTURE NO 38

1-A graphical relationship among different trophic structures or levels in an ecosystem is called the..... **Ecological pyramid.**

2-An ecological pyramid can be divided into main types..... **3.**

3- Determines the flow of energy through each trophic level in a food chain is called the..... **pyramid of the energy.**

LECTURE NO 39

1-A food pyramid is a graphical representation of flow of energy through different trophic levels of an..... **Ecosystem.**

2- The highest level an organism occupies on a food chain is termed as..... **Trophic level.**

LECTURE NO 40

1-Actual place of an organism where it lives is called it's..... **Habitator environment.**

2- Animals always live within a certain range of values, called the.....
Tolerance range.

3- A certain range of values within the tolerance range is called the.....
Range of optimum.

4- When an abiotic factor becomes out of an organism tolerance range then it is said to be a.....
Limiting factor.

5-The example of the limiting factors are.....
oxygen, water & food.

6- Response of an organism towards any abiotic factor is known as.....
taxis.

LECTURE NO 41

1-The sequence of organisms through which energy moves in an ecosystem is called a.....
Food chain.

2- A food chain determines how chemical energy and nutrients move from one organism to another through the trophic levels in an.....
ecosystem.

LECTURE NO 42

1-The most organism form a complex network of the interconnected food chain is called.....
food web.

LECTURE NO 43

1-Producers (autotrophs) obtain nutrition (complex organic compounds) from inorganic materials as an
energy source.

LECTURE NO 44

1-Autotrophs are divided into how many types.....
2

2- Most producers capture sunlight to produce energy rich Carbohydrates (such as glucose) by photosynthesis, which is the way energy enters most ecosystems are called the.....photoautotrophs.

3- Bacteria, can convert simple inorganic compounds from their environment into more complex nutrient compounds by using hydrogen sulphide instead of sunlight through a process called.....chemosynthesis.

4- Nitrifying bacteria are chemosynthetic in nature.....Nitosomnas and Nitrobacter.

LECTURE NO 45

1-On land, most producers are green plants, which generally capture about of the solar energy.....1%.

LECTURE NO 46

1-Organisms that manufacture food in aquatic environment are called the.....aquatic producers.

2- In freshwater and marine ecosystems, algae and aquatic plants are the majorProducers.

3- In open water, the dominant producers arephytoplankton.

4- Microscopic organisms that float or drift in the water are called the.....phytoplankton.

5-How many types of the phytoplankton2(diatoms, dinoflagellates).

6- Too much nutrients causes harmful algal blooms (HABs) which are extremely toxic to living creatures (fish, shellfish, birds and mammals) is called theeutrophication.

LECTURE NO 47

1-Photosynthesis is a process that starts an ecologicalfoodweb.

2- Chlorophyll is the captures light energy directly from sun.....plant pigment.

3- Glucose is then used as a basic source by the primary consumers that feed on plants.....energy source.

LECTURE NO 48

1-Organisms that depends on producers for purpose of food are called theconsumers.

2-There are how many types of the consumers.....4.

3-Animals that feeds on both herbivores & carnivores are called the.....omnivores.

4-Example of the primary consumers iscow ,zebra, rabbits ,grasshoppers , & dears.

5-Example of the 2ndry consumers are Spiders, hyenas, birds, frogs, and some zooplankton-eating fish.

6-Example of the tertiary consumer are Tigers, wolves, mice-eating snakes, hawks, and killer whales.

7-Example of the omnivores is Human pigs, foxes, cockroaches, common carp and tilapia.

LECTURE NO 49

1-Herbivores eat about of plants each day.....150ponds.

2-A giraffe eat about how many ponds each day.....75 ponds.

LECTURE NO 50

1-An animal that preys on and eats a primary consumer. They are also called as..... Carnivores (Meat eaters).

2- Carnivorous fishes possess I stomach and intestine.....
Long,short.

LECTURE NO 54

1-“di-TRI-tus” meaningdebris.

2- Detritus feeders, or detritivores, feed on the wastes or dead bodies of other organisms, which is called asdetritus.

3-Examples of the detritivores are..... Mites, Earthworms
,Someinsects, Catfish.

4-Example of the large scavenger organism.....vultures.

LECTURE NO 55

1-Earth’s life-support system consists of how many spherical systems.....4.

2-The amount of the nitrogen present in the atmosphere is.....78%.

3-The amount of the oxygen present in the atmosphere accounts for21%

4- Most of this water is in the oceans, which cover about of the globe.....71%.

LECTURE NO 56

1-Study of how these characteristics of populations change in response to changes in environmental conditions is called.....**Population dynamics.**

2-Generally randomly distributed populations are rare while clumped distribution is..... **Common.**

3-Ability of a population to grow under ideal conditions.....**Biotic potential.**

4-Animal populations change over time as a result of**Birth, death & dispersal.**

5-Which individuals show high biotic potential.....**bacteria & insects.**

6-The rate at which the population of a species would grow if it had unlimited resources is called.....**Intrinsic rate of increase (r).**

7-The constraints that climate, food, space, and other environmental factors place on a population is called.....**environmental resistance.**

8-biotic potential and environmental resistance determine the.....**Carrying capacity (K).**

9-A population increases by And decreases by
(Birth & immigration), (death & emigration).

10-Apply to populations in which individuals are likely to live out their potential life span.....**Type I curves.**

11-Apply to populations in which mortality rates are constant throughout age classes.....**Type II curves.**

12-Curves apply to populations in which mortality rates are highest for the youngest cohorts.....**Type III.**

13-Group of individuals having same characters is calledcohorts.

14-A population's age structure has a strong effect on..... population growth.

LECTURE NO 58

1-Factors that influence carrying capacity are..... Food, water availability, water, space & environmental conditions.

LECTURE NO 59

1-A rapid increase in number of individuals of a population at a constant rate of growth is called..... Exponential or geometric growth.

LECTURE NO 60

1-Decrease in growth of rapidly growing populations due to limited resources, until they reach their carrying capacity is termed as.....logistic population growth.

LECTURE NO 61

1-Study of human population and the factors that regulate the growth of population is called.....demography.

2-Extinction and colonization are the main process that regulate the..... Metapopulation.

3-A group of populations of same species which are separated by space. Individuals can move from one population to other is called.....metapopulation.

LECTURE NO 62

1-The number of individuals in a population found in a particular area is called.....Population density.

LECTURE NO 63

1-Factors that influence the number of animals in a population and their effect is not dependent on the density of the population are called the.....density independent factors.

LECTURE NO 66

1-The mixture of gases that surrounds the solid earth is called the..... Atmosphere.

2-Raw material for photosynthesis Water vapor.

3-Water vapor plays an extremely important role in thehydrological cycle.

4-97%of this water is saltwater present in.....oceans.

LECTURE NO 69

1-The study of freshwater is known aslimnology.

2-Those water bodies that hold running water are termed as.....lotic water bodies.

LECTURE NO 70

3-A small, shallow and fast moving water body that supports variety of aquatic flora and fauna is called.....stream.

LECTURE NO 72

1-Standing a body of freshwater that supports variety of life is called.....lentic water bodies.

2-Examples of lentic water bodies are.....ponds, lakes & wetlands.

3-The mixing of water from the bottom of the lake with the water close to the surface of lake is called.....**overtturn.**

LECTURE NO 73

1-Deep lakes normally consist of four distinct zones.....**Littoral, Limnetic, Profundal, and Benthic.**

2-A shallow water area near the shore where maximum light can penetrate is called.....**littoral zone.**

3-Most abundant organisms are microscopic phytoplankton and zooplankton..... **Profundal zone.**

LECTURE NO 74

1-Lakes that have a small supply of plant nutrients are called.....**oligotrophic lakes.**

LECTURE NO 75

1-A lake with a large supply of nutrients needed by producers is called a.....**eutrophic lake.**

2-Human inputs of nutrients from the atmosphere and from nearby urban and agricultural areas can enhance the biological activity of lakes by a process called**cultural eutrophication or simply eutrophication.**

3-Lake having excessive amounts of dissolved nutrients is called.....**Hypertrophic lake.**

4-A lake having moderate amounts of dissolved nutrients.....**Mesotrophic lake.**

5-The surface of the eutrophic lake is covered with mats ofalgae & cyanobacteria.

LECTURE NO 76

1-The wetlands which are covered with freshwater all or part of the time and located away from coastal areas are called the.....inland wetlands.

LECTURE NO 78

1-A semi-enclosed body of mixed water where salt water from the sea mixes with fresh water from the land. That mixture of water is known as.....estuarine water or brackish water.

2-The types of the Estuary.....Bar built estuary, Fjords, Tectonic estuary & Coastal plain estuary.

3-A type of estuary that is formed in river valleys due to the action of valley glaciers is called & mostly found in.....fjords ,Norway.

LECTURE NO 79

1-A productive waterlogged area present between land and open sea is known as.....coastal wetland.

2-Some of the earth's most productive ecosystems are known as the.....Mangrove forest.

3-The tangled roots and dense vegetation of mangrove forests act like to reduce damage from storms.....shock absorber.

4-Sea grass beds support at least species that grow underwater.....60%.

5-Marine aquatic systems are a great source of.....biodiversity.

LECTURE NO 81

- 1-Marine life is found in how many zones.....3.
- 2-The coastal zone make up less than % of the world's ocean.....10%.
- 3-The coastal zone contains how % of species.....90%.
- 4-The part of ocean that is deep and away from the coastal zone is termed as..... open ocean.
- 5-The open ocean is divided into how many vertical zones.....3(Euphotic zone, Bathyal zone & Abyssal zone).
- 6-Phytoplankton carry out about % of the world's photosynthetic activity.....40%.
- 7-Showers of dead and decaying organisms drifting downward from upper lighted levels of the ocean are known as..... marine snow.
- 8-Organisms like oysters, clams, and sponges are known as thefilter feeders.

LECTURE NO 82

- 1-The overall economic benefit from marine coastal ecosystems is over \$ trillion per year.....12.

LECTURE NO 83

- 1-A microscopic swimming and free floating organisms in water are termed as..... planktons.
- 2-A very small drifting plankton having size less than 2 micrometer are called as..... ultraplankton.

3-Small photosynthetic bacteria that carry out % of the primary productivity near the ocean surface.....70%.

4-Important examples of nektons are..... fish, turtles and whales.

LECTURE NO 84

1-Corals are tiny living animals that belong to group of organisms called..... Cnidarian.

2-Coral reefs are produced by the large mass of that is secreted by the corals in marine water..... Calcium Carbonate.

3-How many types of the corals.....2(soft & hard).

4-Coral reefs are an amazing centre of aquaticbiodiversity.

LECTURE NO 85

1-The area of shoreline between low and high tides is called the..... intertidal zone.

LECTURE NO 86

1-Weathering is an important phenomenon in forming process.....SOIL.

2-Process of decomposition and disintegration that causes the rocks to break down in a natural way is calledweathering.

3-In cold climates mechanical weathering is.....dominant.

4-In tropical climates chemical weathering isdominant.

LECTURE NO 87

1-In general soils are classified into two main groups on the base of climate.....Pedalfer soils& Pedocal soils.

2-Acidic soils formed in temperate (wet) climates..... Pedalfers soils.

3- The term Pedalfer is a combination of ped, al, and fer
..... Soil, aluminium & iron.

4- Soils that are formed in dry (arid and semiarid) climates and are basic in nature..... Pedocal soils.

5- Pedocal soil sped for , cal for soil & calcium carbonate.

6- A special type of Pedocal that is bright red and forms in tropical and subtropical climates with heavy rain and high temperature is called.....
Laterite soil.

7- A tough cemented layer developed by the dense mass of calcium carbonate is called..... Caliche.

LECTURE NO 89

1-The normal body temperature of humans, goat and birds is.....
98.6°F, 103°F, 105°F.

2- Animals that maintain their internal body temperature constant are called the Warm blooded or the homeotherms.

3-Example of the homeotherms..... birds & animals.

4- Animals whose internal body temperature changes with respect to external environmental temperature called cold blooded or the Poikilotherms.

5-The examples of the cold blooded snake , fishes & frogs.

LECTURE NO 90

1- Ability of an organism to regulate its body temperature is called
..... thermoregulation.

2- In high temperatures the three dimensional structure of enzymes may..... denature.

VU Medical Zone effort by Amaan Khan